

Cheap Signals, Costly Proof: Award-Layer Triage in Cartel Enforcement

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Abstract

Cartel cases begin under thin information. Agencies often see award records long before they recover the bid-level evidence needed for proof. We ask whether that cheap layer can guide where costly forensics should start. Using São Paulo’s BEC procurement platform (2009–2019), we build a *frequent-loser* screen from award records alone: among zero-win firms, participation intensity ranks loser-side cartel-adjacency risk. Computed from 2009–2016 participation, the screen prospectively discriminates adjudicated CADE cobidders in 2017–2019. On the same target, it matches a seven-feature Imhof–Wallimann bid-distribution pipeline and adds non-redundant signal in combination. Used as a gatekeeper, the award-layer screen cuts the bid-microdata pool for the forensic stage by 83% while still flagging 131 of 193 adjudicated cobidders. The statistic creates suspicion, not proof: it ranks cartel-adjacent firms and environments for forensic attention without adjudicating membership or estimating damages.

Keywords: screening under incomplete observability, cover bidding, cartel adjacency, award-layer enforcement, separating equilibrium, participation-based statistics

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1. Introduction

Cartel enforcement begins before cartel proof exists. Agencies must decide where to spend investigative resources while the evidence needed for liability is still costly to recover. That sequencing problem is familiar in the economics of law enforcement: when observation is costly, the first screen should use information already in the agency’s hands, and the expensive evidentiary stage should be reserved for the candidates that screen ranks highest (Becker, 1968; Stigler, 1970; Baker, 2003; Harrington, 2008). Procurement-cartel detection usually begins one layer later. The empirical industrial-organization literature reads coordination off the distribution of submitted bids (Porter and Zona, 1993; Bajari and Ye, 2003; Imhof, 2019; Wallimann et al., 2023; Chassang and Ortner, 2022). Those methods are powerful, but they presume the bid layer: per-bidder bid amounts, within-tender dispersion, bid rankings, and sometimes bid histories. Many enforcement environments routinely expose only a thinner layer: who participated, who won, what was bought, and at what price. The bid-by-bid record may exist, but it sits in operational files that have to be recovered through a case-specific forensic request.

The question here is what antitrust enforcement can do at that earlier layer. The answer is not cartel proof. It is evidentiary triage: a cheap award-layer ranking of firms and procurement environments that indicates where bid-layer forensics should begin. We study this architecture on Sao Paulo’s BEC procurement platform (2009–2019, 1,654,401 tender-items). The screen uses four award-record fields only: participant identity, winner identity, item code, and negotiated price. It flags *frequent losers*: firms with zero wins whose participation intensity among always-loser firms lies above a median plus 1.5 interquartile-range rule. The logic is simple. In a cover-bidding arrangement, loser-side firms are deployed to create the appearance of competition while avoiding the win. The wins-zero filter isolates the side of the bidding pool the award layer can plausibly see; participation intensity within that stratum ranks exposure to the cartel’s deployment problem. The binary frequent-loser rule is a coarsening of that rank, not a claim of legal membership.

That scope determines the validation target. We do not ask whether award records

identify cartel members for sanction. They do not contain the evidence for that claim. Instead, we validate against loser-side cartel adjacency: firms in the always-loser stratum that bid alongside direct defendants in adjudicated CADE procurement-cartel cases. Direct defendants and cobidders play different roles in the evidence. Direct defendants are the membership labels in the legal file and are often winner-side firms. Cobidders are the loser-side footprint the award-layer screen is meant to rank. Stronger discrimination against cobidders than against direct defendants is therefore a design signature, not a fallback target.

Five empirical implications organize the exercise. A useful award-layer screen should discriminate adjudicated cobidders inside the always-loser population, including when the score is formed before the labels are observed. Because the statistic is a loser-side screen, it should perform worse against direct CADE defendants than against cobidders. If cobidders are more than active losing firms, their firm-level and bid-level profiles should resemble credible cover bidding: deployment near direct defendants, concentrated market portfolios, plausibly close losing bids, and dispersion consistent with rotation across cover roles. If award records and bid records carry different information, the award-layer statistic should complement rather than replace bid-distribution methods. Finally, if the architecture has enforcement value, placing the award-layer screen before bid-layer forensics should reduce the bid-microdata pool while preserving most recall against adjudicated cobidders.

The results line up with those implications. Built on 2009–2016 participation and evaluated on 2017–2019 labels, the screen prospectively flags adjudicated cobidders at firm-level AUC 0.864. On the same cartel-adjacency target, it reaches discrimination comparable to a seven-feature Imhof–Wallimann bid-distribution pipeline and adds non-redundant signal when the two are combined (+0.035 AUC, DeLong $p = 0.014$). In the sequential gatekeeper exercise, the award-layer screen reduces the bid-microdata pool for the forensic stage by 83% while still recovering 131 of 193 adjudicated cobidders. The point is not that a four-field award-layer statistic dominates bid-distribution forensics. It does not. The point is that the two layers are complements, and the cheap layer can tell the agency where the costly layer should be applied first.

The cobidder evidence also narrows the interpretation. Within the frequent-loser stratum, cobidders face more unique winners, bid more often alongside direct CADE defendants, and concentrate activity in narrower item-group portfolios than other frequent losers. The LANCES export lets us look one layer deeper: cobidders bid closer to winners and show higher within-firm dispersion in the bid gap, with both signs surviving a multivariate logit that controls for participation intensity. Those facts do not prove cartel membership. They do address the most direct objection to the validation exercise. Cobidders are not merely high-volume losing firms; their loser-side profile is consistent with credible cover bidding in the modern cover-bidding literature ([Asker, 2010](#); [Marshall and Marx, 2012](#)).

The price evidence has a narrower role. Frequent-loser presence is associated with a $+3.6\% - +7.7\%$ higher conditional log negotiated unit price across four broad-sample estimators, and the positive imprint is concentrated in the highest tender-value segment. Under overlap restriction the aggregate sign reverses: three lower tender-value segments are negative and the highest segment remains positive. We report this as economic corroboration that the screen tracks environments where collusion-relevant behavior matters, not as a damages estimate or a causal effect of cover bidding on prices. The main claim does not rest on either sign of the price coefficient.

The contribution is to change where several familiar tools enter the enforcement sequence. For enforcement-design work, we provide an empirical architecture for staged information acquisition under incomplete observability. For bid-distribution screening, it supplies a first-stage award-layer ranking that indicates where bid-distribution moments should be recovered, rather than a substitute for those moments. For bid-coordination tests, it offers an ex ante firm-level partition computable before a case record defines suspected colluders and competitive fringe. For cover-bidding studies, which usually identify cover bidders ex post through leniency or judicial records ([Porter and Zona, 1999](#); [Pesendorfer, 2000](#); [Asker, 2010](#)), it offers prospective loser-side candidates from public award records alone.

Three boundaries matter throughout. The screen creates suspicion, not legal proof. The validation target is cartel-adjacent cobidding, not cartel membership. The price

imprint is descriptive, not a damages calculation. These limits are not cosmetic disclaimers; they define the stage of enforcement studied here. Award-layer triage is useful precisely because it occurs before bid-layer evidence is recovered and before legal proof can be assembled.

The resulting claim is deliberately narrower than cartel detection and broader than a new statistic. When proof is costly and suspicion can be generated from records the agency already holds, the first policy question is not whether the cheap layer can decide liability. It cannot. The question is whether that layer can discipline the allocation of costly forensic attention. In this setting it can. The award layer supplies a legally limited but empirically informative triage stage; the bid layer remains where proof is assembled.

2. Costly Proof and Observable Awards

2.1. Screening, Investigation, and Proof

Antitrust agencies rarely begin with the evidence they need at the end. They begin with coarse records, complaints, leniency leads, procurement anomalies, or repeated interactions that make some firms and markets worth a closer look. Legal proof of a procurement cartel comes later. It combines case records, bid histories, communications, leniency testimony, and economic evidence into a file that can support adjudication. The organizational problem is therefore not only how to classify conduct once rich evidence is available. It is also how to decide which files should receive the costly evidence-gathering effort in the first place.

That distinction changes what a screen is asked to do. Bid-distribution methods ask whether the submitted bids look suspicious, conditional on access to per-bidder bid amounts. An award-layer screen asks an earlier question: before recovering those bid amounts, can the agency rank the firms and procurement environments where bid-level forensics is most likely to pay off? The first exercise is closer to proof; the second is triage. Our interest is in the second exercise. Bid-distribution forensics remains indispensable at the evidentiary stage, but award records may help decide where that stage should start.

2.2. The BEC Enforcement Environment

São Paulo’s BEC platform contains both sides of this information problem. It is a large electronic procurement environment, covering 1,654,401 tender-items during 2009–2019, and it preserves a public contract-award envelope for each tender-item. It also sits inside a legal environment in which cartel cases are adjudicated by CADE, the Brazilian competition authority, creating an external source of legal labels against which an award-layer screen can be evaluated.

Two procurement modalities dominate the sample. *Convite* is a sealed-bid invitation procedure under Lei 8.666/93 with a statutory three-bidder quorum and a low value cap. *Pregão* is a reverse electronic auction under Lei 10.520/2002 in which bidders submit progressively lower offers in real time, with no minimum-bidder rule. These modalities differ in both bidding rule and observability regime. *Convite* is closer to an award-only environment from the analyst’s perspective; *pregão* generates a richer real-time interaction footprint. We use this contrast later as scope information for the screen, not as a source of causal identification. The common element is the award layer that survives across modalities.

2.3. Award Layer and Forensic-Recoverable Bid Layer

BEC exposes two analytically distinct layers. The *award layer* carries the fields an enforcement analyst can query routinely: participant identity, winner identity, item code, negotiated price, procuring unit, year, and procurement modality. The *forensic-recoverable bid layer* carries per-bidder bid amounts, bid timestamps, and bid-revision sequences. Those bid records exist in the BEC operational system, but they are not exported to the ordinary analytical warehouse. They must be recovered through an administrative LANCES export: feasible for evaluation and forensic use, but costly relative to querying the award layer. OECD reviews of Brazilian procurement infrastructure describe this kind of administrative latency as a practical constraint on routine monitoring (OECD, 2019).

Figure 1 summarizes the distinction. The left side reports the bid-layer features required by the Imhof–Wallimann-style forensic benchmark: within-tender coefficient of

variation, dispersion across tenders, skewness, kurtosis, normalized spread, log min–max, and second-lowest bid distance. The right side reports the four fields used by the award-layer screen. Our screen uses only the right side. The bid-layer features enter as the forensic-stage benchmark and as a bid-level validation bridge after the LANCES export has been recovered.

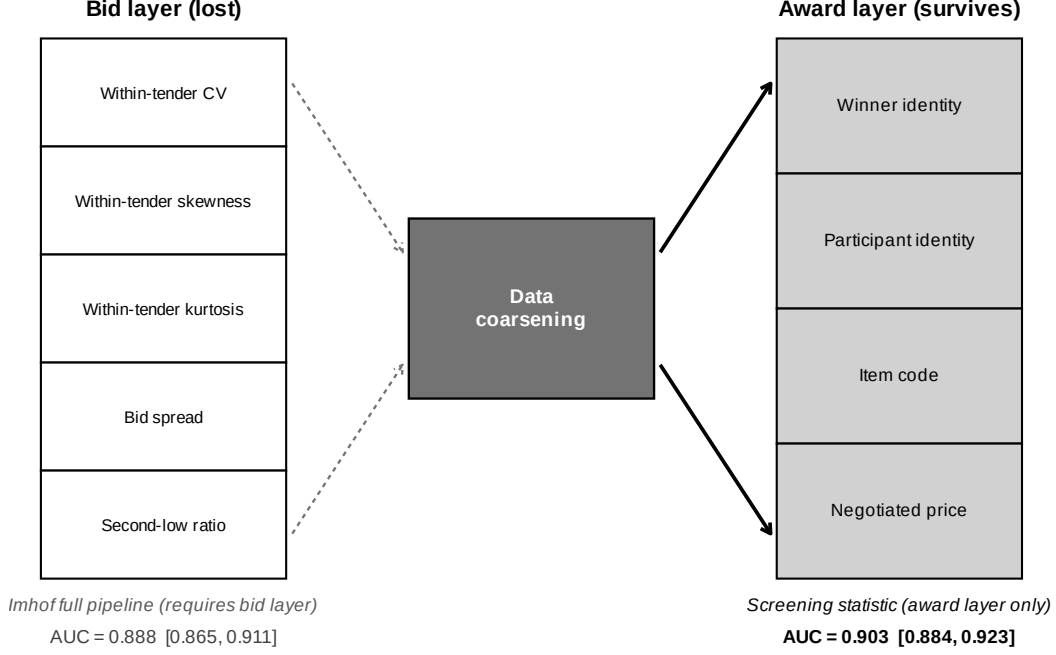


Figure 1: Information layers in procurement-cartel enforcement. Bid-distribution forensics requires per-bidder bid amounts and within-tender bid moments; the award-layer screen uses participant identity, winner identity, item code, and negotiated price. The empirical question is whether the right-hand layer can rank where the left-hand layer should be recovered first.

The two-layer problem is not peculiar to BEC. In the United States, FPDS-NG preserves federal contract awards as routine administrative records, while bid files sit with contracting agencies and are typically recovered through case-specific process. In the European Union, TED mandates publication of contract-award notices, but per-bidder bid amounts are preserved heterogeneously across member-state systems; a recent procurement screening review reports that fewer than half of surveyed systems make bid amounts retrievable at the analytical-warehouse layer (Sánchez Graells, 2019). These examples are not used as external validation. They establish the portability of the information problem: award records are often routine, bid microdata often forensic.

2.4. CADE Labels and the Cartel-Adjacency Target

The legal labels come from CADE’s procurement-cartel adjudications. CADE enforces Brazilian competition law under Lei 12.529/2011, and its leniency and adjudication records identify direct defendants in procurement-cartel cases. The portfolio linked to BEC contains 12 adjudicated procurement cartel cases with 65 firm-defendants, of which 47 are active in BEC during the sample window. Final decisions span 2015–2025, with process-to-ruling intervals of 4–16 years. That lag is central to the screening interpretation. An award-layer screen is valuable only if it can move suspicion toward forensic investigation before the adjudication record exists.

The adjudication record creates two empirical targets. Direct CADE defendants are the membership labels in the legal file, but they are not the natural target of a loser-side award-layer screen. They are often designated winners or otherwise winner-side firms in the cartel’s allocation. Cobidders are different: always-loser firms that appear alongside direct defendants in BEC tender-items. They are indirect labels, and we do not treat them as proof of membership. They are, however, the cartel-adjacency object the award layer can plausibly recover: the loser-side participation footprint adjacent to adjudicated cartel members.

The rest of the analysis follows that boundary. We evaluate the screen primarily against cobidders inside the always-loser stratum. Weak performance against direct defendants is not a failure of the design; it is the expected consequence of a statistic built to rank loser-side exposure. CADE adjudications are used in a legally limited way: as external evidence that some bidding environments were cartel-affected and as direct-defendant anchors for constructing cartel-adjacent cobidder labels. The screen remains a triage device. It does not adjudicate liability.

3. A Loser-Side Screen for Cartel Adjacency

3.1. The Participation Primitive

The award layer cannot observe whether a losing bid was sincere, strategic, or coordinated. It can observe something more basic: which firms repeatedly entered tenders

and never won. That is the primitive used here. The screen begins with the zero-win subset and then ranks firms inside that subset by participation intensity. In the BEC data this conditioning set contains 16,843 always-loser firms. A firm above the participation cutoff is a frequent loser.

The economic reason for starting on the loser side is simple. Cover bidding is useful to a cartel because it supplies the appearance of competition while leaving the designated winner in place. A cover bidder is therefore expected to lose in the cartel-targeted tenders where it is deployed. Zero wins alone is not enough, of course. Many firms lose because they are small, inexperienced, poorly matched to the item, or simply unlucky. The additional signal is persistence: among firms that never win, repeated participation is harder to rationalize as ordinary competitive entry, and easier to reconcile with a role that pays off outside the observed award outcome. The formal appendix states this as a monotone-ranking result: conditional on zero wins, $\log(1 + \text{tenders_count})$ ranks the posterior probability of the cover-bidder type under a sparse separating-equilibrium framework.

This argument is intentionally modest. It does not say every frequent loser is a cover bidder, nor that cover bidding is the only reason for repeated zero-win participation. Capacity signaling, subcontracting relationships, eligibility maintenance, and market-intelligence gathering are also possible. The claim is narrower: the award layer supplies a ranking statistic for loser-side cartel adjacency, and that statistic is worth validating against adjudicated cartel records before any membership language is used.

3.2. The Binary Rule

The continuous score is $\log(1 + \text{tenders_count})$ computed among always-loser firms. For a discrete screen, we use the median plus $1.5 \times$ interquartile range of the always-loser participation distribution. The resulting cutoff is 14 tender-items with zero wins and flags 2,735 frequent losers. The rule is fixed from the participation distribution, before using CADE labels, and requires no bid amounts, bid rankings, or within-tender dispersion measures.

The binary cutoff is useful because enforcement programs often need a list, not only

a rank. Still, the rank is the economic object. The binary rule compresses a continuous participation gradient into a manageable first-stage flag. Later sections therefore report both objects where possible: the frequent-loser indicator as the administrable screen, and $\log(1 + \text{tenders_count})$ as the underlying ranking statistic. This distinction matters in two places. First, a sharp cutoff is easier for firms to manipulate than a rank, a point we return to when discussing adversarial adaptation. Second, bid-distribution forensics can be sequenced after either object: the agency may request bid microdata for all firms above the cutoff, or for the top-ranked firms under the continuous score.

3.3. Populations and Legal-Economic Role

Table 1 fixes the populations used in the analysis. The broad BEC universe contains all firms that appear as participants during the sample window. The screen operates inside the always-loser stratum. Frequent losers are the flagged subset of that stratum. CADE direct defendants are firms named in adjudicated procurement-cartel cases and active in BEC. Cobidders are always-loser firms that appear alongside those direct defendants in BEC tender-items.

Table 1: Populations and legal-economic role

Population	Size	Role in the analysis
All BEC participants	41,444	Firms participating in BEC tenders during 2009–2019; the outer procurement universe.
Always-losers	16,843	Firms with zero wins; the loser-side stratum within which the award-layer screen ranks firms.
Frequent losers	2,735	Always-losers above the participation cutoff; the discrete first-stage screen.
CADE direct defendants in BEC	47	Firms named in CADE procurement-cartel rulings and active in BEC; the legal membership anchors.
Cobidders in the always-loser stratum	193	Always-losers that participated alongside direct defendants; the cartel-adjacency target for validation.

The table also clarifies what the screen is not designed to do. Direct defendants are the legal membership anchors, but they are not the natural target of a loser-side award-layer statistic. Cobidders are indirect labels and cannot adjudicate liability. They

are the validation object because they occupy the part of the bidding pool the screen can observe: firms that lose, keep showing up, and appear near adjudicated cartel defendants. The next section asks whether the frequent-loser ranking actually concentrates on that population at rates above what participation volume alone would imply.

4. Validation Against Adjudicated Cartel Adjacency

4.1. A Conservative Adjudication Benchmark

The cleanest validation uses only cartel cases already adjudicated before the end of 2020. This leaves 4 CADE cases, 30 firm-defendants active in BEC, and 210 always-loser firms that bid alongside at least one of those defendants. Of those cobidders, 108 are frequent losers. Because both the cartel finding and the defendant list precede the construction of the screen, this benchmark avoids relying on later adjudications to define the ground truth.

The main concern is mechanical: firms that participate often are more likely to appear near anyone, including CADE defendants. The benchmark therefore uses a participation-stratified permutation rather than a raw co-bidding rate. Within quartiles of participation volume, we randomly permute cobidder status 1,000 times and recompute the frequent-loser share among the relabeled cobidders. The observed co-bidding rate is 3.95%, against a permutation mean of 1.24%, an excess ratio of 3.2 $\times$ with $p < 0.001$. The ROC AUC against the same conservative ground truth is 0.748 (95% DeLong CI [0.713, 0.783]). The result is not that the screen proves membership. It is that frequent losers are concentrated near adjudicated cartel defendants at rates that participation volume alone does not explain.

4.2. Prospective Discrimination

The broader CADE portfolio is more informative but less clean: it includes cases adjudicated after the BEC sample closes. We use it as a prospective benchmark. The question is whether participation observed before the test window ranks cobidders whose cartel adjacency is revealed only later by the adjudication record. Computing the score from 2009–2016 participation and evaluating it on 2017–2019 labels gives firm-level AUC

0.864 (95% CI [0.858, 0.870]). In-sample AUC is higher, 0.924, as expected; the holdout estimate is the better reference for an enforcement screen that would be run before the case file is complete.

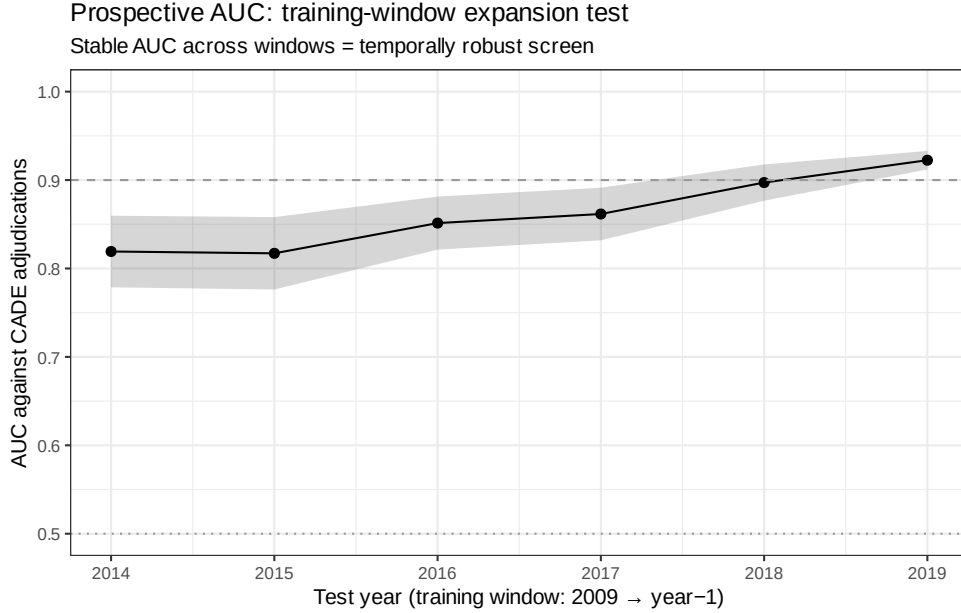


Figure 2: Temporal-holdout ROC curves. Scores are computed using participation observed before the test year; the test year supplies the out-of-time cobidder label.

Figure 2 shows the rolling-origin ROC curves. The point of the figure is not the smoothness of any single year, but the separation from the diagonal across the holdout exercise. A participation statistic constructed before the labels are observed still ranks the loser-side firms later revealed to be adjacent to adjudicated cartel defendants.

4.3. Participation Volume and Leakage

The validation would be uninteresting if it only recovered its own construction. We therefore separate the structural signal from two mechanical components. The first is ordinary exposure: active firms appear in more tenders and have more chances to co-bid with any defendant. The stratified permutation above handles that margin. The second is leakage in item-level classification. If the same cobidder firms help construct the participation score and define the label, in-sample item-level AUC becomes almost tautological; the reference value is 0.995.

Two audits reduce that concern. Holding cobidder firms out of score construction

within cross-validation folds leaves AUC 0.891 (95% CI [0.887, 0.894]). Computing the score only on 2009–2016 participation and predicting 2017–2019 items leaves AUC 0.864 at the firm level. The drop from the in-sample reference is real and should be disclosed; it is the mechanical component. What remains is the part that matters for the screening interpretation: the participation primitive continues to rank cartel-adjacent loser-side firms when the labels are held out of score construction or placed out of time.

4.4. *The Direct-Defendant Asymmetry*

The final validation check runs in the opposite direction. If the award-layer statistic were a general cartel-membership detector, it should also classify direct CADE defendants. It does not. Against direct defendants in the broader BEC firm universe, AUC is 0.49, essentially random. This is the expected result for a loser-side screen. Direct defendants are the firms named in the legal file and are often on the winner side of the arrangement; the statistic is built from zero wins and repeated loser-side participation. A screen designed to rank that side of the bidding pool should not recover the membership object symmetrically.

Table 2: Validation benchmarks

Benchmark	Question	Result
Conservative CADE cases	Are frequent losers overrepresented among pre-2020 adjudicated cobidders, net of participation volume?	3.2× excess; AUC 0.748
Temporal holdout	Does pre-window participation rank later cartel-adjacent cobidders?	AUC 0.864
Leakage audit	Does signal survive when cobidders are held out of score construction?	Out-of-fold AUC 0.891
Direct-defendant asymmetry	Does the loser-side screen classify legal defendants symmetrically?	Direct-defendant AUC 0.49

Table 2 gives the compact view. The four rows are meant to be read together. The conservative benchmark anchors the label in adjudicated cases; the temporal holdout addresses retrospective fitting; the leakage audit owns the mechanical component; and the direct-defendant result keeps the legal object straight. Taken together, they support a limited but useful claim: award records can rank cartel-adjacent loser-side firms before

bid-level proof is recovered. They do not adjudicate membership, and they are not asked to.

5. From Legal Labels to Economic Type

The validation section treats cobidders as the loser-side cartel-adjacency target. That is the right legal object for an award-layer screen, but it leaves an economic question. Are cobidders simply frequent losers who happened to appear near CADE defendants, or do they look like the type of firm a cover-bidding screen is meant to find? We read this question within the frequent-loser stratum, comparing cobidders with frequent losers that do not co-bid with direct defendants. This is the hard comparison: both groups lose persistently and both cross the same participation threshold.

5.1. Firm-Level Signature

Cobidders differ from other frequent losers in the expected directions on most firm-level dimensions. They participate more often, face more unique winners, bid more often in tenders that include a direct CADE defendant, and concentrate activity in narrower item-group portfolios. The unique-winner contrast is especially large: cobidders face 24.8 unique winners on average, compared with 13.5 for frequent-loser non-cobidders (Cohen's $d = +1.00$). The proximity measure also moves in the right direction: 1.5% of cobidders' loss-bids face a direct CADE defendant, compared with 0.2% for other frequent losers ($d = +0.46$). Their portfolios are more concentrated as well, with item-group HHI 0.380 against 0.288 ($d = +0.39$).

One dimension goes the wrong way. Cobidders have a lower share of own loss-bids in repeat winner-pair clusters than frequent-loser non-cobidders (21.6% versus 33.4%). We do not use that measure as support for the cover-bidding interpretation. The broader firm-level profile is still informative: the cobidder label is not just a proximity tag attached to otherwise identical active losers.

5.2. Bid-Level Signature

The LANCES export lets us check whether the same distinction appears inside bids. For firms with at least five usable losing bids in *convite* or *pregão*, we compute

the gap between each losing bid and the winning bid in the same tender-item, $(\text{bid} - \text{winning bid})/\text{winning bid}$, winsorized at $[-0.99, 10]$ before firm-level aggregation. The usable sample contains 182 cobidders and 2,369 frequent-loser non-cobidders.

The bid-level pattern is not the old textbook version of cover bidding, in which cover bids are plainly uncompetitive. Cobidders bid closer to winners, not farther away: their median bid gap is 0.582, compared with 0.809 for other frequent losers ($d = -0.281$). At the same time, their within-firm dispersion in bid gaps is higher (1.207 versus 1.099, $d = +0.147$). A logit of cobidder status within the frequent-loser stratum, controlling for $\log(1 + \text{tenders_count})$, preserves both signs: the median-gap coefficient is -0.708 ($z = -4.14$), and the dispersion coefficient is $+0.442$ ($z = +3.62$). The bid-level distinction is therefore not only another way of measuring high participation.

Table 3: Cobidder signature within the frequent-loser stratum

Dimension	Cobidders	FL non-cobidders	Difference
Total participations	136.5	76.7	$d = +0.67$
Unique winners faced	24.8	13.5	$d = +1.00$
Loss-bids facing direct CADE defendant	1.5%	0.2%	$d = +0.46$
Portfolio item-group HHI	0.380	0.288	$d = +0.39$
Repeat winner-pair share	21.6%	33.4%	$d = -0.38$
Median bid gap to winner	0.582	0.809	$d = -0.281$
Within-firm SD of bid gap	1.207	1.099	$d = +0.147$

5.3. What the Signature Establishes

The pattern is most consistent with credible cover bidding. Cover bids must lose, but they need not be absurdly high. In many cartels they must remain plausible enough to preserve the appearance of rivalry and to avoid drawing attention ([Asker, 2010](#); [Marshall and Marx, 2012](#)). A firm that cycles through cover roles may therefore bid relatively close to the winner in some tenders while showing elevated dispersion across tenders. That is the pattern we observe.

This is not a proof of cartel membership. The LANCES evidence is used for type validation, not adjudication. It narrows the most important alternative reading of the

validation results: that cobidders are merely high-volume losers mechanically adjacent to CADE defendants. They are high-volume losers, but they also have a firm-level and bid-level profile that fits the economic type the award-layer screen is meant to rank. With that target in place, the next question is whether the award-layer screen can organize the more expensive bid-layer tools used in cartel forensics.

6. Sequencing Screening and Forensic Proof

The previous sections establish the target and the type. The remaining question is institutional: what should an agency do with a signal that is cheap enough to compute before it has the evidence needed for proof? The answer is not to treat the award-layer screen as a substitute for bid analysis. The answer is to put it before bid analysis. In that sequence, suspicion is created on the record layer the agency already observes, and forensic work is reserved for a smaller set of firms and environments where the expected return to costly bid-level inspection is higher.

6.1. Bid-Layer Benchmark

The natural benchmark is the bid-distribution tradition in cartel screening. Imhof–Wallimann-style detectors use within-tender bid moments, such as dispersion, skewness, spread, and distance from the winning bid, to detect patterns consistent with coordinated bidding (Imhof, 2019; Wallimann et al., 2023). Those moments live one layer deeper than the frequent-loser screen. They require per-bid offer values and enough usable bid microdata to construct a firm-level forensic profile. The award-layer screen requires only participation counts and the absence of wins.

Table 4 puts the two layers on the same cartel-adjacency target. Within the always-loser pool for which bid-distribution features can be computed ($N = 16,779$), the binary frequent-loser flag reaches AUC 0.903 (95% CI [0.884, 0.923]). The continuous participation score reaches AUC 0.884 ([0.860, 0.908]). The full bid-layer pipeline reaches AUC 0.888 ([0.865, 0.911]). This is not an outperformance claim. The point is sharper: a record-layer statistic, computed before forensic recovery of bid values, discriminates

the same adjudication-anchored target in the same range as a much richer bid-layer benchmark.

Table 4: Discrimination at Different Information Costs

Model	Information required	AUC	95% CI
Frequent-loser flag	Award records: participation count and zero wins	0.903	[0.884, 0.923]
Participation intensity	Award records: $\log(1 + \text{tenders_count})$ and zero wins	0.884	[0.860, 0.908]
Bid-layer benchmark	Per-bid offer values and within-tender bid moments	0.888	[0.865, 0.911]
Award layer + bid layer	Participation intensity plus bid-distribution moments	0.962	[0.954, 0.969]

Notes: Five-fold cross-validated AUC against CADE-cobidder labels within the always-loser pool for which the bid-distribution benchmark can be computed ($N = 16,779$). The combined specification uses the same target and adds the award-layer score to the bid-layer moments.

6.2. Complementary Signals

Comparable AUCs do not by themselves justify a two-stage architecture. If the award-layer statistic were just a noisy version of the bid-layer score, it would be useful only where bid microdata are missing. The stronger result is that the two layers are not redundant. Adding the award-layer statistic to the bid-layer benchmark raises discrimination to 0.962 ([0.954, 0.969]). In the identical same-sample audit, where all models are evaluated only on firms with complete bid-layer features, the frequent-loser flag adds +0.035 AUC over the full Imhof–Wallimann benchmark alone (DeLong $p = 0.014$). The combined specifications gain +0.096 — + 0.098 AUC over the bid-layer benchmark ($p < 0.001$).

That gain is substantively important for JLEO purposes because it separates two legal-economic objects. Bid moments are closer to forensic proof: they speak to the shape of bidding inside a tender. Participation intensity is closer to investigative priority: it asks whether a firm repeatedly supplies losing participation in markets where winners rotate through the award layer. A regulator should not collapse those objects. The first creates a reason to look; the second helps evaluate what is found after looking. Their combination improves prediction because the two stages measure different parts of the suspected coordination process.

6.3. A Sequential Gatekeeper

The operational test is not whether the cheap screen beats the forensic benchmark. The relevant test is whether it can reduce the forensic workload without throwing away too many adjudication-anchored positives. Table 5 reports that exercise for 11,676 always-loser firms with bid-layer features and 193 CADE-cobidder positives. Joint scoring, which gives the classifier both the award-layer score and bid-layer moments for every firm, is the upper bound for an agency willing to recover bid microdata on the full pool. At the top 1,000 flags, it finds 142 cobidders, or recall 0.736.

The sequential design asks for less. It first ranks firms using the award layer. It keeps the top $K_1 = 2,000$ firms. Only then does it recover or process bid microdata and apply the bid-layer benchmark within that survivor pool. At the top 1,000 final flags, the sequential rule finds 131 of 193 cobidders, or recall 0.679. The loss relative to joint scoring is 8%, while the bid-microdata footprint falls by 83%.

Table 5: Gatekeeping the Bid-Layer Forensic Stage

Rule	k	TP	Precision	Recall	Bid-microdata firms
Award-layer only	500	66	0.132	0.342	0
Bid-layer only	500	73	0.146	0.378	11,676
Joint scoring, award + bid	500	112	0.224	0.580	11,676
Sequential: award $\rightarrow$ bid, $K_1 = 2,000$	500	96	0.192	0.497	2,000
Joint scoring, award + bid	1,000	142	0.142	0.736	11,676
Sequential: award $\rightarrow$ bid, $K_1 = 2,000$	1,000	131	0.131	0.679	2,000

Notes: Evaluation pool: 11,676 always-loser firms with computable bid-distribution features; positives are 193 CADE-cobidders. The bid-microdata column reports how many firms require per-bid offer values before the rule can be applied.

The sequential rule is not the precision champion; joint scoring has the easier job because it uses every information layer for every firm. The sequential rule earns its place in a different way. It makes the expensive forensic stage conditional on a cheap suspicion stage, and the empirical cost of doing so is modest relative to the reduction in bid-microdata exposure. That is the tradeoff faced by an enforcement agency that cannot inspect every bidder, every bid, and every tender with the same forensic intensity.

6.4. *Temporal Audit*

The in-sample gatekeeper exercise still deserves skepticism. A classifier trained and evaluated inside the full sample may rank firms well because it sees a long history close to the adjudicated outcome. We repeat the gatekeeper audit with features restricted to the pre-target window, 2009–2016, while evaluating against the same post-sample adjudication-anchored cobidder target. The temporal-holdout pool contains 8,257 always-loser firms with complete train-window bid features and 142 positives.

The temporal audit preserves the architecture. At top 500, the award-only rule loses 47% of its in-sample true positives, joint scoring loses 24%, and the sequential rule loses only 9%. In levels, the sequential rule recovers 87 positives at top 500, compared with 85 for joint scoring. At top 1,000, it recovers 114 positives, compared with 111 under joint scoring. The result is not that temporal holdout eliminates all fragility. It does not. The result is narrower and more useful: conditioning bid-layer forensics on an earlier award-layer screen remains competitive out of time while keeping the forensic footprint small.

The legal interpretation follows the same boundary as the empirical design. A frequent-loser flag cannot establish an agreement, identify a damages base, or prove that a particular bid was cover. It can identify where the agency should spend the next unit of investigative effort. The bid-layer stage then examines whether the bidding record contains the richer patterns needed for proof. For a journal concerned with how law works under institutional constraint, that sequencing is the contribution: a cheap administrative record can improve the allocation of costly legal evidence gathering without pretending to be legal evidence itself.

7. **Economic Harm, Legal Scope, and Limits**

The enforcement value of the screen does not depend on proving a price effect. A screen can be useful if it ranks where forensic attention should go, even when the price record is too coarsened to support damages or causal incidence. Still, a cartel-adjacent screen should leave some trace in the economic outcome agencies care about. In BEC,

frequent-loser presence is associated with higher negotiated unit prices in the broad sample, but the association changes sign under overlap restrictions. That tension is informative. It says the price evidence should corroborate the screening interpretation, not carry the main claim.

7.1. Price Imprint

In the broad item-level price regressions, frequent-loser presence is positively associated with log negotiated unit price. The coefficient is 0.0677*** with item and year fixed effects, 0.0636*** after adding procuring-unit fixed effects, 0.0933*** in pregão, and 0.0382** in convite. Across the main descriptive estimators, the implied range is +3.6% – +7.7%. The pregão–convite split matters for interpretation. If frequent losers were mainly quorum fillers created by the three-bidder rule in convite, the price imprint should be stronger there. Instead, the larger estimate appears in pregão, the real-time electronic environment where loser-side positioning is easier to deploy repeatedly.

Table 6: Price Imprint and Scope

Exercise	Estimate	Standard error / p	Interpretation
Broad sample, item and year FE	0.0677***	0.0230	Positive price imprint
Broad sample, item, year, PBU FE	0.0636***	0.0215	Positive after buyer controls
Pregão subsample	0.0933***	0.0255	Larger in electronic auctions
Convite subsample	0.0382**	0.0186	Smaller under quorum-prone procedure
Overlap-cell ATT	−0.097	$p < 0.001$	Aggregate sign reversal
PS-trimmed ATT	−0.307	—	Stronger negative under trimming
Tender-value Q4, overlap ATT	+0.041	$p = 0.045$	Positive in high-value segment
Direct-CADE overlap items	−0.061	$p = 0.45$	Statistically null

Notes: Dependent variable is log negotiated unit price. The broad-sample rows use the main BEC item-level sample. The overlap rows restrict or reweight comparisons to cells with treated and untreated support. The estimates are descriptive and are not used as damages calculations.

The positive broad-sample association is not a causal estimate of cover bidding’s effect on prices. The available quasi-experimental routes do not support that claim: discontinuity designs around procedure thresholds lack the required density behavior, and the policy-change design does not supply credible parallel trends. For the argument here, the price evidence has a smaller job. It asks whether the places highlighted by the screen also look economically relevant. On the broad record they do.

7.2. Where the Price Claim Stops

The overlap results draw the boundary. When comparisons are restricted to cells where frequent-loser-present and frequent-loser-absent items overlap on observables, the aggregate coefficient reverses to -0.097 ; the PS-trimmed estimate is -0.307 . Only 1.06% of treated items lack a within-cell counterfactual, so the reversal is not mainly a dropped-observation story. It is a reweighting story: the broad estimate and the overlap estimate answer different questions about different parts of the item-value distribution.

The segment decomposition makes that point explicit. In the lowest three tender-value quartiles, the overlap estimates are negative. In the highest tender-value quartile, the estimate is positive: $+0.041$ ($p = 0.045$). That is the segment where cartel rents are most plausible ex ante because the same fixed cost of coordination is spread over larger contracts. The same table also shows a null estimate inside direct-CADE overlap items, -0.061 ($p = 0.45$), which is another warning against reading the price regressions as a clean treatment effect.

The mixed price record disciplines the legal claim. The evidence says that the award-layer screen ranks cartel-adjacent environments; it does not say that every flagged item is overpriced, that the observed coefficient is a markup, or that the screen identifies a damages base. Those are different legal tasks and require different evidence.

7.3. Portability and Strategic Response

The screen is portable only where the institutional primitives exist. A candidate jurisdiction needs public or auditable award records with participant identities, winner identities, item codes, and procurement timing. It also needs a procurement environment in which persistent losing is economically meaningful rather than purely mechanical. BEC fits that profile particularly well: standardized goods are common, electronic records are available, and pregão-style auctions preserve the loser-side interaction environment in which the signal is strongest. Settings dominated by bespoke services, infrastructure, or sparse one-off procurement may still be screenable, but the item-level price comparison and the frequent-loser threshold would need to be rebuilt for the new setting.

Strategic adaptation is the other limit. The screen is fairly stable to simple rotation of cover bidders (0.931 versus baseline 0.928) and to occasional wins that evade a strict zero-win rule (0.929). It weakens under CNPJ splitting (0.891) and degrades more under threshold-aware capping (0.801). Combining adaptations lowers AUC to 0.632. That pattern is not surprising. A visible threshold invites threshold gaming. In practice, the rule should be recalibrated, monitored for bunching near the cutoff, and paired with bid-layer forensics where those data can be recovered.

7.4. Conclusion

Cartel enforcement often begins before the agency has the evidence it would need to win a case. Procurement records then pose a practical question: can the cheap layer do any legally useful work before the expensive layer is opened? In São Paulo’s BEC platform, the answer is yes, but only within a disciplined scope. Persistent zero-win participation ranks firms that are cartel-adjacent on the loser side. It predicts adjudicated cobidder exposure out of time, lines up with an economic cover-bidder signature, and helps decide where bid-distribution forensics should be run.

The contribution is not a shortcut to liability. It is a way to order suspicion under incomplete information. That distinction is why the screen belongs in the enforcement architecture rather than in the proof stage. Used alone, it gives an agency a cheap triage list. Used before bid-layer forensics, it reduces the microdata footprint while preserving most of the adjudication signal that a full joint model would recover. Used beyond that, it would be overclaimed. The statistic identifies where legal attention should start, not where legal responsibility ends.

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